(7) $\log\left|\log\left(\tan\frac{x}{2}\right)\right|+c$ (8) $\frac{1}{b^2}\log|a^2+b^2\sin^2 x|+c$ (9) $\frac{(\sin^{-1}x)^2}{2}+c$

(10) $(1+\sqrt{x})^2-4(1+\sqrt{x})+2\log\left|1+\sqrt{x}\right|+c$ (11) $\log|\log(\log x)|+c$

(12) $-e^{-\beta x^{\alpha}}+c$ (13) $2\sqrt{\sec x}+c$ (14) $\frac{(1-x)^{19}}{19}-\frac{(1-x)^{18}}{18}+c$

(15) $\frac{\sin^6 x}{6}-\frac{\sin^8 x}{8}+c$ (16) $(x-a)\cos a-\sin a\log|\sec(x-a)|+c$

Exercise 11.7

(1) (i) $e^{3x}[3x-1]+c$ (ii) $-\frac{x\cos 3x}{3}+\frac{\sin 3x}{9}+c$

(iii) $-e^{-5x}[5x+1]+c$ (iv) $x\sec x-\log|\sec x+\tan x|+c$

(2) (i) $\frac{x^2\log|x|}{2}-\frac{x^2}{4}+c$ (ii) $e^{3x}[9x^2-6x+2]+c$

(iii) $x^2\sin x+2x\cos x-2\sin x+c$ (iv) $-x^3\cos x+3x^2\sin x+6x\cos x-6\sin x+c$

(3) (i) $-\sin^{-1}x\sqrt{1-x^2}+x+c$ (ii) $\frac{1}{2}e^{x^2}[x^4-2x^2+2]+c$

(iii) $\frac{1}{2}\left[4x\tan^{-1}4x-\log\left|\sqrt{1+16x^2}\right|\right]+c$ (iv) $2\left[x\tan^{-1}x-\log\left|\sqrt{1+x^2}\right|\right]+c$

Exercise 11.8

(1) (i) $\frac{e^{ax}}{a^2+b^2}[a\cos bx+b\sin bx]+c$ (ii) $\frac{e^{2x}}{5}[2\sin x-\cos x]+c$

(iii) $\frac{e^{-x}}{5}[2\sin 2x-\cos 2x]+c$

(2) (i) $-\frac{e^{-3x}}{13}[3\sin 2x+2\cos 2x]+c$ (ii) $-\frac{e^{-4x}}{10}[2\sin 2x+\cos 2x]+c$

(iii) $\frac{e^{-3x}}{10}[\sin x-3\cos x]+c$

Exercise 11.9

(1) $e^x\log|\sec x|+c$ (2) $\frac{e^x}{2x}+c$ (3) $e^x\sec x+c$

(4) $e^x\tan x+c$ (5) $xe^{\tan^{-1}x}+c$ (6) $\frac{x}{1+\log|x|}+c$

13_11th-Maths-Vol2-Answers.indd 278 16-12-2022 11:13:19